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INNOVATIONS IN PROMPT ENGINEERING FOR IMPROVED DATA PROCESSING AND ANALYSIS

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ABSTRACT

Designing effective prompts for the aim of getting valuable outputs from artificial intelligence systems is the focus of the emerging field of prompt engineering, which is focused with the design of prompts. The development of artificial intelligence has resulted in the emergence of a demand for trained quick engineers who are capable of maximising interactions between humans and AI, particularly in the field of natural language processing. After conducting research and conducting analysis, the author of this essay comes to the conclusion that India possesses the necessary resources to develop educational programmes and cultivate talent in order to become a global leader in rapid engineering. There is a creative potential in the interactions that occur between computer science, psychology, language, and prompt engineering. Even though it is present on a global scale, businesses are beginning to recognise the potential of artificial intelligence (AI) when it is provided with the appropriate guidance. This has resulted in an expected annual growth of over twenty percent in the need for quick engineering. The country of India is in an excellent position to capitalise on this opportunity because it possesses a large pool of qualified engineers. The development of huge language models is receiving a lot of attention from a lot of people, and they are making significant progress in this area. One of the primary causes is the introduction of OpenAI's ChatGPT models, specifically the GPT3.5-turbo and the GPT-4. A innovative approach to the production of synthetic data and the distillation of knowledge is presented in this article through the utilisation of fast engineering. The three approaches that are the primary focus of our attention are the fundamental, the composite, and the similarity approaches. One of the most common challenges that machine learning applications face is dealing with imbalanced datasets, and the objective of this research is to investigate the capabilities of various algorithms to deal with such datasets. When compared to the entire dataset, not a single one of the strategies that are based on prompts performs adequately in the studies. On the other hand, the method of similarity prompting is superior to the other approaches and has a great deal of growth potential. According to the findings, there appears to be a significant opportunity to enhance these technologies and provide synthetic data that has a greater diversity of variables.

Keywords: Data Processing, Prompt Engineering, AI, ChatGPT

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INTRODUCTION

The activity of building and optimising prompts to elicit specific and relevant results out of artificial intelligence systems, particularly those that use natural language processing (NLP), is contributing to the development of a new multidisciplinary field that is being named "prompt engineering" [1]. A prompt is the form of input that an artificial intelligence system gets. This prompt provides the machine with some background information as well as specific instructions on how to reply. Over the past few years, there has been a significant rise in the demand for specialists who are able to train large language models such as GPT-3 to generate text, graphics, and other forms of media that are reminiscent of human writing. In order to handle realistic difficulties and use cases that occur in the real world, these models are now undergoing training.

Although "prompt engineering" is a relatively new term, it is founded on ideas that have surfaced over the course of the past ten years because of advancements in natural language processing and innovations in pretrained language models. According to the findings of the researchers [2], deliberate cues, as opposed to simpler ones, allowed artificial intelligence systems to provide responses that were more coherent and meaningful. Because of this, research was conducted to determine the most effective approach to design prompts in order to maximise the effectiveness of AI for a variety of applications. The term "prompt engineering" refers to the process of combining concepts learned from the fields of languages, psychology, and communication with those learned from computer science (for example, programming and machine learning).

This is the first phase in prompt engineering, which involves constructing a prompt to limit the problem area to something that the model can reliably manage. The first step in prompt engineering is to understand the capabilities and constraints of an artificial intelligence system. It is necessary for prompt engineers to consider nuances in language, emotion, reasoning, and human psychology to effectively direct the artificial intelligence. To preparing the model for the structure and substance of the answer that is anticipated, it is necessary to provide essential background regarding the work or scenario and to adequately clarify the instructions. Using templates, it is possible to achieve the goal of standardising prompts for typical use cases [3]. Engineering can be considered both an art and a science because it is a field that makes use of creative language abilities in conjunction with empirical testing.

Companies have come to the realisation that there is a need for specialised rapid engineers to optimise the deployment of artificial intelligence for jobs such as customer service, content generation, data analysis, targeted suggestions, and other similar activities. Because of this, there has been a significant increase in the demand for engineers like these. A wide variety of industries, including technology, consulting, publishing, healthcare, and banking, have seen a rapid increase in the number of job opportunities available to engineers. Prompt engineers collaborate closely with computer scientists, business leaders, and several other stakeholders to guarantee the smooth incorporation of artificial intelligence into processes and products. Artificial intelligence systems can provide valuable outcomes when they are given well-designed prompts; nonetheless, their results are frequently damaging or irrational. The field of engineering is rapidly becoming a subject that is taught in educational institutions because it is becoming increasingly important and encompasses a wide range of disciplines. The course work includes a variety of topics, including the fundamentals of language, ethics in artificial intelligence prompting, the relationships between machine learning and psychology, logic,

computer programming, and practical laboratories to increase prompt testing abilities. In addition, there have been worldwide conferences that have been established specifically for the goal of displaying innovative engineering techniques and the latest technological advancements. The 2022 International Prompt Engineering Challenge [4] is one piece of evidence that demonstrates that prompt engineering is a distinct field of research.

In recent months, one field of artificial intelligence that has seen great growth is natural language processing (NLP), which stands for natural language processing. Particularly, there has been a lot of attention paid to generative language models like GPT (Generative Pretrained Transformers), which are examples of such models. The reason for this is that ChatGPT has extraordinary text-generating capabilities that come close to imitating human discourse. These capabilities are made possible by ChatGPT's simple access to content that is in the public domain. The effectiveness of these models has been astonishingly shown in a wide range of applications, including sentiment analysis, summarization, translation, and the generation of writing. But the production of these intricate models requires a significant number of resources. There is an absolute requirement for the collection and storage of vast volumes of data, in addition to a significant amount of computing power. When one considers the impact on the environment as well as the high financial expenses associated with power consumption and data storage, the necessity for extensive computational resources and storage capacity presents several significant and challenging obstacles. Expenses of this nature are out of reach for all organisations, except for the most well-known ones, which makes the research process more difficult.

Within this unique environment, the events of this article take place. We study several approaches to producing data based on prompts by utilising the GPT-3.5-turbo language model that is available from OpenAI. ChatGPT is an extension of GPT that includes capabilities and features that are specifically designed for conversational encounters. To accomplish our mission of providing helpful insights and methods, we will be investigating the possibility of prompts. We will be able to make greater use of these models and make them more useful in a variety of contexts because of this.

LITERATURE REVIEW

As a result of recent developments in artificial intelligence (AI) and natural language processing (NLP), there has been a lot of activity in the field of research concerning large language models (LLM) in relation to chatbot technology. Chatbots and artificial intelligence have been steadily improving over the course of recent years. As a result of their evolution, which is based on LLM training on vast datasets, they are now able to deliver replies that are comparable to those that people provide [5]. GPT (generative pre-trained transformer) series large-scale language models [6] have been introduced, and they are displaying remarkable performance across a variety of natural language processing applications. This is a noteworthy development.

Generative artificial intelligence [6] is a strategy that requires the learning of content patterns and depends on this method as a basic component to generate new content using inference results. OpenAI, Google, Deepmind, and Meta are just few of the global technology companies and academic institutions that are currently engaged in several large-scale projects. Typical examples of generic corpus datasets that are used for training models such as the GPT series include web pages, books, essays, and other textual materials. Because it makes use of adaptive methods such as fine-tuning, it can thereafter be used to a variety of natural language processing tasks. due of its millions of parameters, LLM can learn a wide variety of language patterns and structures. This is due of the number of parameters it has. Not only is this LLM technical innovation creating ripples in the realm of artificial intelligence, but it is also altering the game for the development and utilisation of AI algorithms.There is a possibility that the pre-trained

language model (PLM) could have its modelsize or data size expanded [7–10]. This would be done to increase the LLM's modelling capabilities.

For example, larger PLMs such as GPT-3 (175B parameters) and PaLM (540B parameters) have recently been trained to gain a better understanding of performance boundaries. These enormous PLMs outperform the smaller ones, which are represented by the 330M parameter BERT and the 1.5B parameter GPT-2, in terms of performance. This is the case even though the expansion is normally carried out at a variety of model sizes using the same architecture and pre-training activities. Language models that have been trained on enormous volumes of text data, such as GPT-3, PaLM, Galactica, and LLaMA, typically have hundreds of billions of parameters, and as a result, they are referred recognised as large-scale language models (LLMs) [11].

A closer look reveals that ChatGPT is a model that is based on GPT-3.5 and has been updated from the previous GPT-1 to GPT-3 through enhancements to both the model and the learning process. From GPT-1 to GPT-3, the most important change is the modification of the model size. This modification is intended to improve performance by acquiring new information from a variety of datasets. Few-shot learning, which enables efficient model learning with sparse data, and prompt learning, which enables the utilisation of domain knowledge by using human-readable text as input, are the two approaches that GPT-3 employs to train its models [12]. Both methods are necessary for the training of its models.

Just a few of the many tasks that can be accomplished with prompt-based learning include communication, translation, webcoding, and random writing. According to GPT-3.5 [13], it also allows for human interaction during the entire learning process and is fine-tuned to perfection. It was decided to implement the reinforcement learning algorithm known as RLHF, which stands for reinforcement learning from human feedback, to make GPT-3.5 less inherently biassed and harmful. Currently, the RLHF model incorporates human preferences by letting humans rank its responses and provide feedback through a reward function. This is done to include human preferences. [14] The three-stage learning technique makes it possible to acquire further knowledge of GPT-3.5 by utilising the RLHF algorithm and prompt-based supervised learning. To summarise, the three processes are as follows: first, the collection of demo answers and the verification of policy compliance; second, the collection of comparison data and the training of a reward model; and third, the optimisation of policies through the utilisation of a reinforcement learning algorithm [15].

KNOWLEDGE DISTILLATION USING PROMPTS

The complete dataset of IMDB movie reviews is shown in IV-A, and it is utilised in the research that we are conducting. By utilising the GPT-3.5-turbo model that ChatGPT employs, we were able to generate synthetic movie assessments by utilising the official OpenAI application programming interface (API). Because it is often believed that the temperature of ChatGPT's model is set to be somewhere between 0.6 and 0.8, we decided to set it to 0.8 for the model that was chosen. Performing an analysis of the dataset, creating prompts, generating prompts, accessing the application programming interface (API), processing responses, and rebalancing the dataset were all components of the process, as shown in Figure 1.

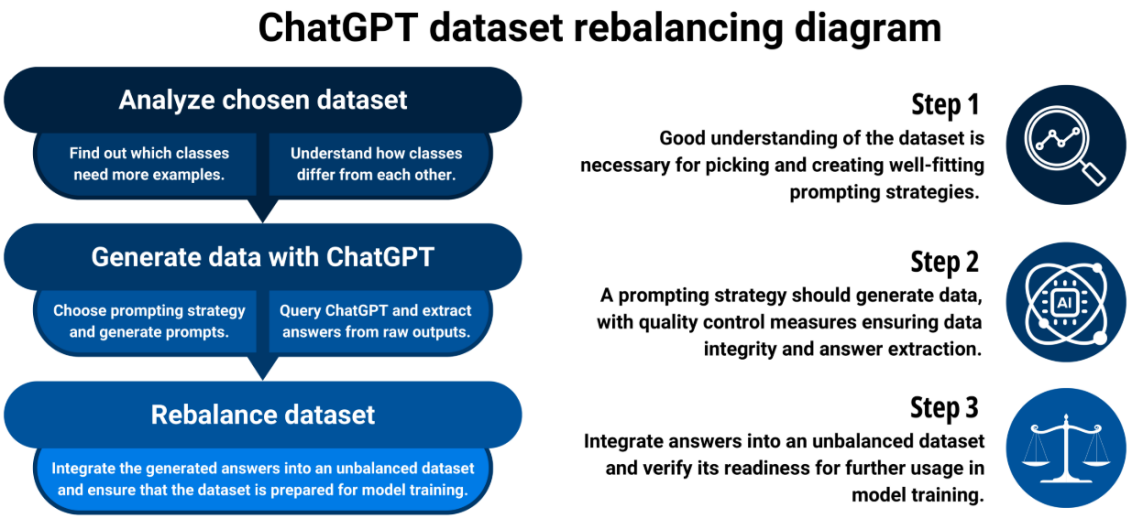

Fig. 1: Flow diagram showing the three stages of dataset rebalancing with ChatGPT: 1) analyzing the dataset and choosing classes for synthetic data generation; 2) choosing prompting strategy, querying (prompting) the ChatGPT service and extracting answers; 3) rebalancing the dataset by integrating extracted answers.

To get the desired results from the model, we painstakingly built the queries as part of the rapid engineering approach. We developed the composite and similarity prompts and stored them to a CSV file during the prompt generation step, or we created the fundamental questions in JSON format. To get the model's answers, we sent requests with prompts using the OpenAI API. Finally, we uploaded the processed and produced synthetic reviews to the dataset. We demonstrated and assessed three separate methods of prompting:

1) Basic prompts A single, straightforward prompt yields three critical analyses of an unnamed film.
2) Composite prompts – The statements and probabilities in Listing 2 are used as prompts to generate a more diversified dataset.
3) Similarity prompts– We are looking for many reviews that are similar to the one in the original dataset.

A. Basic Prompts

We developed a simple prompt to make data generation easy and straightforward. Listing 1 of the JSON file shows that its structure complies with the standards set by the OpenAI API.

Example of negative movie reviews from basic prompt:

Negative review 1:

The special effects were cheap and unconvincing. The dialogue was clichéd, and the pacing was slow. I checked my watch constantly, waiting for the film to end.

Negative review 2:

The acting was terrible, **the special effects were cheap and unconvincing**, and the script was full of plot holes. Save your money and skip this film.

Fig. 2: An example of movie reviews generated by using basic prompt shown in listing 1. We can see that both reviews have the same sentence.

System messages still help configure the assistant's actions, despite claims in the OpenAI documentation that the GPT-3.5-turbo model pays little heed to them. The user message is the main directive that describes the current task and the expected structure of the model's output. Two reviews that were made following basic guidelines are shown in figure 3 and example 1.

Examples of movie reviews:

Positive review:

Julie Andrews and Rock Hudson were great in this movie / musical. The opening song by Ms. Andrews, "Whistling Away the Dark," will always be in the back roads of my mind. The plot line during World War I is great and suspenseful. If you are a romantic, you will love this movie. This is a movie that I have always enjoyed seeing again and again.

Negative review:

This fanciful horror flick has Vincent Price playing a mad magician who realizes his vocational talents have been sold to another. He devises ways of avenging all those that have wronged him. His master scheme seems to backfire on him. Price is a little below par compared to his masterpieces, but it is still the only reason to watch this thriller. The supporting cast includes Patrick O'Neal, Mary Murphy, Eva Gabor, and Jay Novello.

Fig. 3: Example one of both positive and negative reviews from IMDB dataset.

IMPLEMENTING PROMPT ENGINEERING EDUCATION

It is vital to have skills training and individualised curricula that integrate technical knowledge with advanced linguistic and psychological comprehension in order to achieve rapid engineering competency. Using a modular, project-based approach makes it possible to provide engineering education that is both rapid and adaptive.

The fundamentals of computer science, such as data structures, algorithms, programming, and cloud computing, ought to be covered in the beginning courses that come with the subject. Through participation in the required courses, students will acquire a foundational understanding of neural networks, machine learning, and natural language processing technologies. Reinforcement learning, conversational artificial intelligence, and transformer architectures are some of the topics that would be explored. Among the most important pillars are analytics, statistics, and mathematics alike.

Students would learn how to create prompts that are effective in a wide variety of contexts if they took courses in prompt engineering for example. There are a variety of courses that may be taken in this field, including Prompts for Creative Applications, Prompts for Knowledge Applications, Multi-modal Prompting, Responsible and Ethical Prompt Engineering, and Introduction to Prompt Engineering. Language, psychology, and human-centered design are the subjects that are covered in classes that give a sociocultural foundation. Additionally, taking a few courses in engineering technologies and tools might be highly advantageous.

Project-based learning is necessary for achieving practical competency. For students to be successful, they should work swiftly and receive feedback frequently. This is because the difficulty of the optimisation issues they work on grows. It is possible that a project will include prompt optimisation for a variety of reasons, including but not limited to coherence, originality, relevance, tone, factual correctness, and others. The provision of domain specialisations is possible through the implementation of projects that are tailored to industries, such as healthcare, banking, education, retail, government, and others.

The combination of internships and apprenticeships is the most effective method for providing students with real-world experience in a particular subject. More individuals can take part, and more individuals are able to obtain apprenticeships in trades. It is quite valuable to have the opportunity to gain knowledge by forming partnerships with companies that incorporate prompts into their goods and services. Students can observe both the goals of prompt engineering and the challenges that they face in the actual world.

Interdisciplinary academic members who have expertise in social science and artificial intelligence and natural language processing are of the utmost importance. It is feasible to make up for a shortage of professors by engaging in international collaborations, inviting guest speakers, and employing co-teaching strategies. An approach that can be considered is to supplement in-person learning with knowledge that is provided by leaders online. To cultivate engineers who are responsible and thorough, it is necessary to place an emphasis on ethics and responsibility throughout the entire curriculum.

In addition to teaching students the fundamentals of technology, it is essential to equip them with the following transferable skills: the ability to think creatively and analytically, the capacity to work in teams, the ability to comprehend and appreciate different cultures, the ability to communicate effectively, the ability to think critically, the ability to comprehend and empathise with users, the ability to conduct effective research, and the ability to have a business mindset. The objective of technical education in the field of engineering is to produce professionals who can bring about positive change, rather than simply producing technically proficient engineers.

A balance needs to be struck between giving students with a strong foundation in engineering and being adaptable enough to accommodate new technologies, such as artificial intelligence, as it emerges. Education in India needs to find this balance. Students should be provided with good research and self-learning capacities as part of the curriculum for them to maintain their relevance in the modern world. The provision of credentialing and micro-credentialing serves as an excellent method for professional organisations to facilitate learning that continues throughout one's life.

Taking great care in the process of planning and delivering their curricula, educational institutions in India have the potential to rapidly produce graduates who are well-rounded in the subject of engineering and who are prepared to effectively apply their skills in a range of environments. It is essential to place a strong emphasis on both the training of skills and the inculcation of an ethical mindset to actively lead the development of artificial intelligence. If India's educational institutions can generate graduates who are capable of competing on a global scale, then India will move to the top of the list as a rapidly developing centre for engineering talent.

EXPERIMENTAL SETUP

In this section, the experimental procedures that we carried out to gather information for this article are broken down in detail. To speed up the development process, the implementation was written in Python and utilised several different libraries. In this section, we detail the complete methodology of the experiments that were carried out for the purpose of this paper. It is important for the reader to have a comprehensive understanding of the processes that were utilised in the execution of the tests. This will enhance the reader's ability to appreciate the results that will be presented in the coming sections. Figure 4 provides a visual representation of the complete experimental assembly.

Fig. 4: Overview of general experimental setup. It consists of five consecutive steps: selecting the primary dataset, data generation for unbalanced classes using ChatGPT, dataset rebalancing with synthetic data, classification model training, and performance evaluation.

Dataset

The Internet Movie Database (IMDB) movie reviews dataset is a collection of reviews written in English that have been categorised as either positive or negative and taken from the IMDB website. On this website, users are able to rate and evaluate films that they have viewed using a scale that ranges from one to ten. In accordance with this particular dataset, reviews that receive a score of four or lower are considered to be negative, and reviews that receive a score of seven or higher are considered to be positive. We restrict the number of reviews that are published for each film to a maximum of thirty. This is due to the fact that the ratings for the same film tend to be relatively comparable.

There are 50,000 reviews in the dataset that have been labelled, and they are distributed as follows: 25,000 positive reviews and 25,000 negative reviews were received. There are 25,000

records in each of the two datasets, which are the training dataset and the test dataset. There are two distinct types of movie sets that are utilised for training and testing purposes.

Experimental Design

Experiments were divided into three primary categories within this study: those that were carried out on a balanced dataset, those that were carried out on an artificially imbalanced dataset, and those that were carried out on a dataset that was both artificially imbalanced and replenished to a balanced condition using synthetically generated data. In order to verify whether or not the designed methods of prompt-based synthetic data synthesis could be utilised to repair datasets that were skewed, each and every one of these tests was carried out.

When we were putting together each experiment, we did so with a great deal of caution. Our first order of business was to prepare the dataset for training. The initial ten percent, twenty percent, thirty percent, or all of the initial unfavourable review samples, depending on the figure that we choose to select. All of the experiments that were conducted on the unbalanced dataset utilised the same collection of negative reviews that were taken from the training dataset. It is possible that the results of all of the research can be compared more simply using this method. Furthermore, in order to ensure that all of the approaches were evaluated in a reliable manner, we created five balanced stratified folds and one unbalanced stratified fold using the training dataset. Then, for each and every trial, we utilised a five-fold stratified cross-validation model. Furthermore, in order to evaluate the classifiers that were generated by each cross-validation cycle, we utilised the initial IMDB testing dataset.

EXPERIMENTSL RESULTS

BERT

In accordance with its promises, the BERT model, which has a great reputation for its performance on several natural language processing tasks, demonstrated excellent effectiveness across the board. Its outstanding marks on the F1 score, accuracy, and precision metrics are evidence that it is accurate.

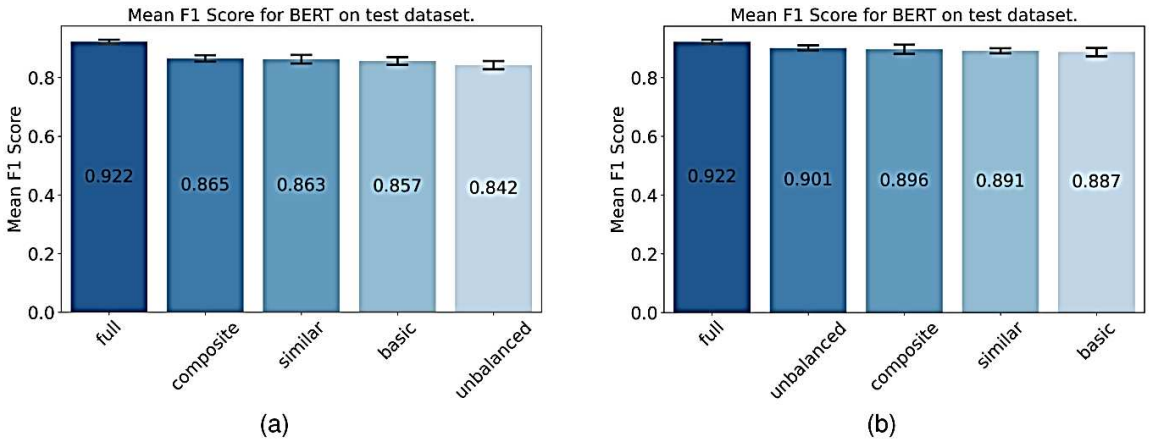

Fig 5: F1 Score for BERT

When examining the tests that were 10% imbalanced, the configuration known as "Composite 10%" comes out as the one that performed the best in terms of the F1 score. This is demonstrated in Figure 5a. The fact that this is the case provides some evidence that composite prompts might be effective in circumstances where there is a significant separation between socioeconomic

classes. Ironically, the imbalanced dataset itself outperforms the other methods in the studies that were 20% unbalanced, which is an intriguing discovery. This discovery gives rise to an intriguing hypothesis: perhaps it is preferable to let the dataset to be imbalanced rather than to generate synthetic data and introduce possible noise beyond a certain point. Although the differences between the various methods are fairly small in both the 10% and 20% imbalanced setups, it is important to note that the composite prompts worked well and appeared to be the better alternative for producing synthetic data when using the BERT model. This is something that should be mentioned.

Random Forest Classifier

The performance of the Random Forest classifier is significantly different from that of the entire dataset when contrasted with the imbalanced experimental settings that were used in the past. The results presented in Figure 6 demonstrate that the Similarity prompts strategy is superior than the other methods when applied to the 10% and 20% imbalanced datasets. Specifically, this is the case with regard to the F1 score.

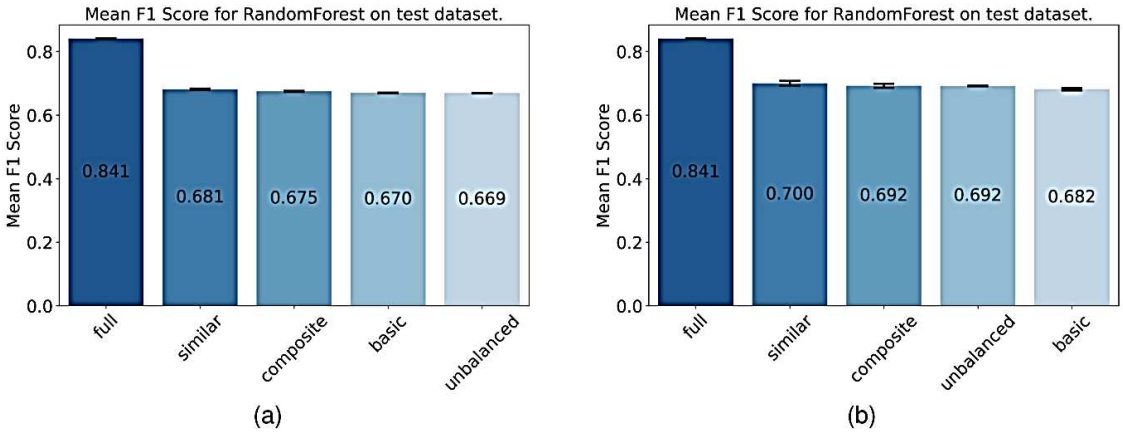

Fig. 6: F1 Score results for experiments with RandomForest classifier. (a) Plot for tests with 10% of train dataset. (b) Plot for tests with 20% of train dataset.

The similarity prompt method is equally as helpful when it comes to accuracy and recall metrics as it is when it comes to other strategies. On the other hand, the fundamental urges strategy has a negative impact on performance in the 20% arrangement, and it seems to have absolutely no impact whatsoever on the outcomes in the 10% setup. These findings indicate that when addressing a class imbalance in a Random Forest model, it may be useful to incorporate equivalent occurrences in the minority class. This is demonstrated by the fact that the findings demonstrate that. The fundamental process, on the other hand, is not only exceedingly inefficient but may even provide even less desirable outcomes.

Naive Bayes Classifier

The Naive Bayes classifier compares favourably to the Random Forest model in terms of its overall performance. This concept is useful with a variety of classifiers, as demonstrated by the fact that the similarity prompts method surpassed all of the other methods in terms of metrics. The basic prompts strategy comes in first place when it comes to performance, while the composite prompts technique comes in second place. The experiment with an imbalanced dataset produced the worst results, which reinforces what we already knew: that the methods used to generate data had a greater influence on performance than the alternative, which was to leave the dataset unbalanced.

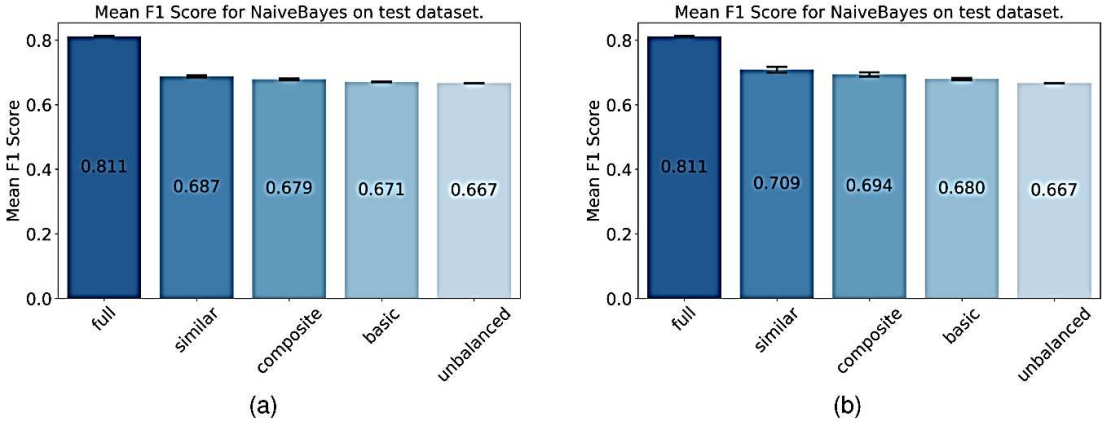

Fig. 7: F1 Score results for experiments with NaiveBayes classifier. (a) Plot for tests with 10% of train dataset. (b) Plot for tests with 20% of train dataset

It was the Naive Bayes classifier that generated the results with the highest degree of reliability. The results obtained from the data collected with similarity prompts were the best across all criteria, with composite prompts coming in second place. The results presented in Figure 7 demonstrate that the balanced dataset fared less well than the basic prompts.

CONCLUSIONS AND FUTURE SCOPE

The basic, composite, and similarity prompt methods for producing synthetic data using the GPT-3.5-turbo model are described in this article. We found that moving from Python list to JSON improved the arrangement of the information when dealing with composite prompts to collect synthetic data. We developed these inquiries bearing this finding in mind. Our goal in conducting these tests was to determine the potential efficacy of supplementing the chosen sentiment analysis dataset with synthetic data. The three approaches were evaluated based on the effectiveness of models trained on this dataset. The one with the highest promise was selected. While running the tests, it became abundantly evident that none of the proposed techniques were advanced enough to produce results on par with the complete dataset. Although it scored 8 to 20 percentage points lower than the whole dataset, the Similarity prompts strategy offered the greatest results. Composite prompts provided a few small advantages, such a simple structure and low cost, but basic prompts were completely ineffective and offered no real benefits. There is obviously not enough variety in the results. The outcomes are still very similar even if the generative model doesn't reproduce the data it produces word for word. When compared to imbalanced datasets, metrics still improve with just those three solutions.

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